

Applications of Artificial Intelligence

Unit 4 Consolidation Task

Task 1: Defining Key Concepts

Agent: An agent is a system or a programme that can collect data from its environment and apply that data to complete self-directed tasks and achieve pre-determined goals. Agents perceive their environments and collect data through sensors, and act upon their environment to complete their tasks through actuators (Norvig & Russell).

Percept: Percepts are the inputs that an agent perceives from its environment. These could include light, water, temperature, or the presence of physical objects.

Actuator: An actuator is a device that converts energy into physical motion. The energy converted could be, for example, electrical, hydraulic or thermal.

In the context of Artificial Intelligence systems, an actuator is the part of the system that physically performs the tasks, typically converting electrical energy to kinetic energy. This can include, for example, robotic manipulators (“arms”)

Rationality: Rationality is the ability of an AI agent to make informed choices, combining data from their environment and past context to achieve optimal performance.

Imitating behaviour involves an AI system learning a process by observing and imitating a human or another AI system. Imitation learning identifies the physical movements required in tasks, for example, in the assembly of products in a factory. Imitating behaviour focuses on the observable outputs of the behaviour concerned.

Imitating thinking involves an AI system observing and demonstrating the thought process of a human or other AI system. It focuses on internal thought processes. It is demonstrated in some Large Language Models (LLMs) that can show the process by which they solve a problem, breaking the problem down into steps, showing intermediate conclusions, and tracking a path to their final response. This mimics the human train of thought.

Imitating thinking places emphasis on thought processes, while imitating behaviour emphasises producing outputs.

Task 2: Analysing Task Environments

(a) Self-driving car in an urban setting

Performance measure: Time taken to reach destination, passenger comfort.

Environment: Road safety, buildings, pedestrian safety, road signs, zebra and pelican crossings.

Actuators: The car itself (physical), particularly the engine, wheels, headlamps and other physical features that interact with the road.

Sensor: Space sensors (to detect cars nearby), and temperature sensors (to help control internal temperature for passenger comfort).

Observability: Partially observable. The car can detect most of the road, but blind spots might be present.

Determinism: Fairly stochastic, as the environment can be uncertain (e.g. unexpected turns by other road users, and weather changes). However, there is also some determinism because the car can control where it drives and when it stops.

Agent interaction: multi-agent, as other actors are present (other drivers, pedestrians, traffic wardens, and passengers).

(b) Virtual personal assistant (e.g., Siri or Alexa)

Performance measure: User feedback, response speed, proportion of commands understood by personal assistant at the first attempt.

Environment: User environment (Home or Office)

Actuators: The users, websites and apps accessed by the virtual assistant.

Sensor: Sound sensors to detect voice commands. Temperature sensors for household temperature control. Light sensors for detecting lighting turning on and off.

Observability: Partially observable because the sensor can detect some aspects of their environment, but cannot detect the human user's thoughts, which would help to understand fully what the user requires.

Determinism: Fairly stochastic, as user's requirements and reactions can change. Also, information available to the virtual PA, for example, through internet searches, will vary. As such, the same actions by the virtual PA will not always yield the same results.

Agent interaction: Multi-agent, as multiple actors exist such as the user, the user's contacts, and organisations providing information that will be accessed by the virtual assistant (such as streaming services, search engines, online maps, and transportation services).

(c) A robotic vacuum cleaner in a home

Performance measure: User feedback, cleanliness of completed floor, speed, safety.

Environment: Household floors

Actuators: The users, dust and dirt (if automatic)

Sensor: Space sensors (detect walls and open spaces)

Observability: Fully observable, as it can be programmed to detect dirt and space, which are the most relevant factors to its operations. However, it is also arguably partially observable because there will be some relevant factors the cleaner will not observe, such as causes of dirt.

Determinism: Stochastic. The future causes and location of dirt are neither known to nor controlled by the cleaner.

Agent interaction: Multi-agent, as operations are also affected by surrounding humans and objects.

Task 3: Agent Design Comparison

	Reactive	Model-based	Goal-based	Utility-based	Learning Agents
Definition	Maps inputs to outputs based on data or training. Has no memory of past interactions	Uses condition-action rules to operate, relying both on settings and experience of its environment.	Adopts a goal-oriented approach, wherein actions are geared towards achieving specific objectives.	Adopt a utility-oriented approach with a focus on maximising satisfaction (which can be measured on a continuous scale).	Update their behaviour based on new experiences and data.
Strengths	Simple to understand and repair.	Adapts to it's environment, holding	Work towards a set goal, while also being	Can work considering multiple factors, and	Are flexible and can adapt to changing

	Work quickly with minimal processing	memory of past interactions.	able to address obstacles and challenges along its path.	effectiveness is measured through different outputs.	environments and users.
Weaknesses	Heavily dependent on fixed settings, as it typically holds no memory and might not be suitable for complex or dynamic situations where the optimal actions should consider on several different factors.	These systems may take time to adapt when moved to a new environment (e.g. a vacuum robot moved to a new house) Model-based systems might be unable to handle complex and dynamic situations.	Can be limited in scope, relying on preset strategies, and unable to respond to all challenges.	These systems can be complex to design and manage. They make decisions using different considerations, and it can be complex to measure the weight to be given to this.	Operate based on data and experiences, and so could produce wrong outputs where source/training data is erroneous or misinterpreted.
Example	Pelican crossings in traffic management. When the user pushes the button, the crossing can set the traffic light to red, and then the pelican crossing light to green. This is done without advanced	A robot vacuum cleaner operates based on its settings as well as stored memory of its environment, such as the location of physical obstacles.	A navigation system that can find directions from one point to another, which uses a fixed map, and is unable to respond to new circumstances like roadblocks and diversions	E-commerce systems, particularly where price discrimination is applied. Goods can be priced based on customer profile, demand levels, inventory levels, seasons, and competitor prices.	Modern AI-powered chatbots with NLP (Natural Language Processing).

	processing or reasoning.				
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(b).

A utility-based system will work well for weather forecasting. Weather forecasts are based on observations (e.g. from satellites and weather stations), as well as mathematical models of airflow and thermodynamics, and past weather patterns. A utility-based model will consider different factors, assign weights to each and create output based on this.

Task 4. Critical Thinking Essay

Intelligent surveillance agents collect and process data about people and environments and can use this to predict movement and social behaviours and make decisions.

Intelligent surveillance agents differ from basic surveillance systems because in addition to collecting data, they can make inferences and decisions in some cases. They include CCTV technology with facial recognition, predictive policing tools, workplace productivity monitoring, as well as social media platform monitoring.

These agents can promote public safety through predictive technology, as with predictive policing and social media monitoring. Surveillance systems are also applied in retail for behavioural analytics, which can be done unobtrusively through ambient intelligence systems. These can track customer movements in retail spaces, measuring interactions with shop displays to inform stocking and marketing decisions.

Despite the valid use cases, there are also negative effects. Surveillance systems trained on historic data can amplify existing biases, especially in predictive policing. Planning based on erroneous or outdated data could cause the misallocation of policing resources.

Additionally, there is the potential invasion of the privacy of individuals. Consumers are often not meaningfully informed of ambient intelligence agents in retail spaces. Legally mandated notices are sometimes buried in the small print of voluminous privacy policies. This raises vital questions about the ethics of these systems.

Further, although these agents might lack autonomy (meaning that final decisions on actions are taken by a human), they perform complex tasks and work with large amounts of data that is difficult to review and understand, so there remains a risk that humans will implement actions suggested by the agents without a proper review of its processes.

Overall, more extensive regulation of surveillance systems is not only ethical, but necessary, and it is desirable for the law to evolve to codify measures that presently only exist as ethical concerns.

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